COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 1 : Session – 11 : SLO – 2 | EX-NOR gate
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II. Fill in the Blanks

• 1. same (equal)
• 2. Y = (A⊕B)■ = AB + A■B■
• 3. ■ (EX-OR with inversion bubble)
• 4. equality (error/parity)
• 5. equality

I. Match the Following

• 1. EX-NOR Logic Expression — b) Y = (A⊕B)■
• 2. Output Condition — a) 1 when inputs are equal
• 3. Symbol Used — c) ■ or ⊕■
• 4. Truth Table Output for (1,0) — d) 0
• 5. Practical Application — e) Digital comparators and equality checking

III. Activity / Describe and Perform

• Logic function: EX-NOR is the complement of EX-OR — output 1 when both inputs are EQUAL. Boolean expression: Y = (A⊕B)■ = AB + A■B■.
• Symbol: EX-OR symbol with an inversion bubble at the output.
• Truth table: 00→1, 01→0, 10→0, 11→1.
• Practical application: digital comparators — one EX-NOR per bit; all bits equal → all outputs 1 → AND them to detect equality of two numbers. Verifying all four input combinations experimentally reproduces the truth table.